



Case Study: Multi-Agent Documentation Automation for a Global Nonprofit

Snapshot

- **Client:** Global humanitarian nonprofit
- **Engagement Lead:** Scott Turnbull, CIO (embedded with the client)
- **Services:** AI agent development, documentation automation, knowledge transfer
- **Timeline:** 4 weeks from pilot to full rollout
- **Highlights:** 12+ codebases documented, onboarding time cut from 3 months to 1 week, avoided vendor lock-in

"When we handed a newly documented codebase to a developer who had never seen it before and they stood it up in hours, I knew the crew was working."

— Scott Turnbull, CTO

The Challenge

The nonprofit delivered data infrastructure and software to humanitarian agencies worldwide, but its vendor-built applications lacked even basic documentation. More than two dozen Python and React codebases, supporting UN agencies, the Red Cross, and national NGOs, were understood only by the original offshore developers, most of whom had already moved on

With the vendor shutting down, Tech Tavern had three weeks to:

- Transition active projects to new development partners without breaking service delivery.
- Satisfy new client audits requiring technical, security, and operational documentation.
- Preserve revenue streams tied to ongoing maintenance and feature delivery.

Manual documentation would have taken an estimated 12 person-months of unfunded effort. Failing to deliver risked losing future contracts, triggering clawbacks on current work, and jeopardizing life-critical humanitarian data services.

The Solution

Scott led a sprint to build a custom multi-agent framework in CrewAI, tapping the then-new Claude 3 API to accelerate the work. The approach unfolded in four phases:

1. **Policy & Standards** — Authored a documentation best-practices policy covering docstrings, architectural overviews, installation guides, and security procedures. This created the acceptance criteria for every output.
2. **Pilot & Calibration** — Prototyped a CrewAI-based agent swarm and validated it on TheDeep.io, an internal platform serving thousands of humanitarian data analysts. Internal stakeholders reviewed the AI-produced documentation to confirm quality and completeness.
3. **Stakeholder Alignment** — Presented the rollout plan and timeline to external partners to secure buy-in, clarify anonymization needs, and align on audit requirements.
4. **Full Production & Hand-Off** — Tuned the multi-agent workflow for scale, ran it across 12 priority codebases, and supplemented edge cases with targeted manual updates. Each cycle concluded with client-ready packets: docstring coverage, architectural maps, setup guides, and security notes.

Because agentic CLI tooling was still nascent, Tech Tavern built custom readers, writers, and memory management utilities to handle large repositories and tighter context windows. The team also paired the Senior Backend Engineer and Senior Data Engineer with Scott to transfer multi-agent skills while validating the outputs.

Results That Matter

By the end of the four-week window, the organization transformed its documentation posture and operating resilience:

- **12+ codebases** fully documented with docstrings, architecture diagrams, and operational runbooks.
- **Onboarding time** for new developers dropped from **~3 months to 1 week**, unlocking faster transitions and reducing unpaid ramp-up by at least 6×.
- **Vendor independence** restored: projects could be reassigned to new agencies or internal teams without risking knowledge loss.
- **Audit readiness** achieved: the nonprofit passed new quality and security reviews, preserving revenue and credibility.
- **Scalability unlocked**: teams could now replicate the documentation playbook across future projects, avoiding the rebuild-from-scratch trap.

As funding tightened across the humanitarian sector in 2025, the comprehensive documentation ensured paused projects could restart without losing institutional knowledge—protecting mission-critical services that underpin aid delivery worldwide.

Key Takeaways for Mission-Driven Teams

- **Treat documentation as an asset:** Build policies before you build tools so “done” has a shared definition.
- **Pilot on a familiar system:** Use a well-known internal project to calibrate AI agents and win stakeholder confidence quickly.
- **Pair automation with skill transfer:** Engage senior engineers in the agent workflow to build internal capability, not just outputs.
- **Attack technical debt in sprints:** Target the highest-impact repositories first, then roll improvements across the portfolio in sequenced waves.

"New tools let us erase years of technical debt in weeks. You still need the right skills and governance, but once you unlock that, the organization becomes agile again."

— Scott Turnbull, CTO

About Tech Tavern

Tech Tavern helps mission-driven organizations harness AI responsibly. From readiness assessments and governance frameworks to agent development and fractional CIO advisory, we turn high-stakes technology initiatives into repeatable successes.

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"Build with Intelligence. Amplify Your Impact."